MobiWeaver: A Multi-Agent Framework for Cross-City Human Mobility Generation

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ACL ARR 2026 May Submission8297 Authors

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May, Senior Area Chairs, Area Chairs, Reviewers, Authors, Secondary Reviewers
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Keywords: LLM Agents, Multi-Agent Framework, Reinforcement Learning, Human Mobility Generation

Abstract:
Human mobility generation aims to synthesize realistic point-of-interest (POI) trajectories for individual users, yet cross-city generation requires a model to capture both transferable mobility regularities and city-specific spatial facts. Existing Large Language Model (LLM) approaches typically encode urban spatial facts into model parameters, incurring costly retraining when transferring to new cities. To address this challenge, we propose MobiWeaver, a hierarchical multi-agent framework for cross-city human mobility generation. MobiWeaver decomposes generation into a city-agnostic planner that predicts coarse itineraries and a tool-augmented searcher that grounds these itineraries to concrete trajectories. To stabilize training, we further introduce Tool-Anchored Credit Policy Optimization (TACPO). TACPO localizes POI-related action tokens and converts rollout-level advantage into action-level credit signals using effectiveness-aware and position-aware advantage rescaling. TACPO protects valuable search actions from downstream failures and emphasizes early decisions that anchor subsequent mobility generation. Experiments on three held-out cities show that MobiWeaver achieves the best performance across five JSD metrics and reduces the average JSD by 16.59 relative to the second-best baseline for each city-metric pair.

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Research Area Keywords: NLP for social good, tool use, llm agents, reinforcement learning
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A2 Potential Risks: Yes
A2 Elaboration: Ethical Considerations
B Use Or Create Scientific Artifacts: Yes
B1 Cite Creators Of Artifacts: Yes
B1 Elaboration: 3
B2 Discuss The License For Artifacts: Yes
B2 Elaboration: 3
B3 Artifact Use Consistent With Intended Use: Yes
B3 Elaboration: 3
B4 Data Contains Personally Identifying Info Or Offensive Content: Yes
B4 Elaboration: Ethical Considerations
B5 Documentation Of Artifacts: Yes
B5 Elaboration: 3

B6 Statistics For Data: Yes
B6 Elaboration: 3
C Computational Experiments: Yes
C1 Model Size And Budget: Yes
C1 Elaboration: appendix A.3
C2 Experimental Setup And Hyperparameters: Yes
C2 Elaboration: appendix A.3
C3 Descriptive Statistics: Yes
C3 Elaboration: 3
C4 Parameters For Packages: Yes
C4 Elaboration: 3
D Human Subjects Including Annotators: No
D1 Instructions Given To Participants: N/A
D2 Recruitment And Payment: N/A
D3 Data Consent: N/A
D4 Ethics Review Board Approval: N/A
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Official Review of Submission8297 by Reviewer uuCd

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Official Review by Reviewer uuCd (weilinruan (/profile?id=~weilinruan1))
08 Jul 2026, 14:23 (modified: 12 Jul 2026, 23:34)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uuCd
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Paper Summary:
This paper proposes MobiWeaver, a hierarchical multi-agent framework for cross-city human mobility generation. The method decomposes generation into a planner, which predicts coarse time-window/category itineraries, and a searcher, which grounds these itineraries into concrete POI trajectories using external retrieval tools. The paper further proposes TACPO, a tool-anchored reinforcement learning method that localizes POI-related action tokens and rescales rollout-level advantages based on action effectiveness and position. Experiments are conducted on three held-out cities from Massive-STEPS, using five JSD-based distributional metrics: SD, SI, DARD, STVD, and ActP. The paper reports that MobiWeaver outperforms deep learning, general LLM, and mobility LLM baselines, and provides ablations for the hierarchy, reward components, and TACPO (Sec. 2-3; Table 1-4).

Summary Of Strengths:

1. The paper studies an important and timely problem: generating realistic human mobility trajectories in cross-city settings, where both transferable behavioral patterns and city-specific POI information matter (Sec. 1).
2. The planner/searcher decomposition is intuitive. Separating coarse itinerary generation from tool-based POI grounding is a reasonable design for reducing city-specific memorization and enabling database-based transfer (Sec. 2.2; Fig. 2).
3. TACPO targets a real issue in tool-based RL training. The observation that rollout-level rewards may assign poor credit to intermediate POI search actions is plausible, and the proposed effectiveness-aware and position-aware rescaling mechanisms are easy to understand (Sec. 2.4.1; Eq. (7)-(9)).
4. The paper includes a relatively broad set of baselines, including deep learning models, general LLMs, and mobility-generation LLMs (Sec. 3.1; Table 1).
5. The ablation studies are useful. The paper separately examines the hierarchy, planner reward components, and TACPO components, which helps identify where the reported gains may come from (Sec. 3.3; Table 2-4).

Summary Of Weaknesses:

1. The main empirical comparison is not fully convincing. MobiWeaver uses a complex pipeline with teacher-generated SFT data, RL training, external retrieval tools, and two specialized 4B policies. Several baselines appear to be used mostly through prompting or city-specific training, and it is unclear whether they receive comparable training data, teacher supervision, RL optimization, or tool-use adaptation (Sec. 3.1; Appendix A.2-A.3). This makes the large gains in Table 1 hard to attribute specifically to the proposed hierarchy or TACPO.
2. The evaluation relies mainly on population-level JSD metrics. SD, SI, DARD, STVD, and ActP measure aggregate distributional similarity, but they do not sufficiently evaluate individual-level trajectory fidelity, exact POI prediction, user-specific consistency, or downstream utility (Sec. 3.1; Appendix A.5). The paper trains with more fine-grained rewards but evaluates mostly with coarse aggregate statistics.
3. The training data construction raises concerns. The SFT data are synthesized by a very large teacher model and filtered by an LLM-as-judge; the RL data also remove cases that the teacher fails to synthesize reliably (Appendix A.2). This may bias the training set toward easier or more teacher-compatible cases, but the effect of this filtering is not analyzed.
4. Reproducibility is limited. The method depends on a self-constructed city database, synthesized user profiles, 58K SFT examples, 50K RL examples, teacher model outputs, LLM-as-judge filtering, and large-scale H200 training (Appendix A.2-A.3). The paper says the resulting data and database will be released in the future, but the current submission does not provide enough information to independently reproduce the full pipeline.
5. The baseline setup is underspecified. Several baseline model names and versions are very recent or nonstandard, and the paper does not provide exact checkpoints, prompts, decoding settings, sampling parameters, or cost budgets for each LLM baseline (Sec. 3.1; Table 1). This is especially important because many baselines are LLM-based and sensitive to prompting/tool formatting.
6. The ablations are averaged over three cities without uncertainty estimates. Table 2-4 do not report variance, confidence intervals, or significance tests. Given the relatively small held-out city set and multiple metrics, it is hard to judge whether the reported differences are robust.
7. The ethical discussion is not sufficient for a mobility generation paper. Human mobility data are highly sensitive, and the paper synthesizes user profiles and generates personalized POI trajectories. The ethical section states that identifiers and residential addresses were removed, but it does not discuss re-identification risk, misuse of generated mobility traces, or safeguards for releasing the generated data/database in enough detail (Ethical Considerations; Appendix A.2).
8. The NLP contribution is somewhat indirect. The work uses LLM agents and natural-language user profiles, but the core evaluation is an urban mobility generation benchmark. The paper would be stronger if it more clearly explained what is learned about language-based agents beyond this specific urban computing application.

Comments Suggestions And Typos:

1. Add a fairness table comparing MobiWeaver and each baseline in terms of training data, tool access, SFT/RL adaptation, teacher supervision, number of parameters, and inference budget.
2. Add individual-level metrics, such as exact POI hit rate, category accuracy, time error, sequence edit distance, or user-conditioned likelihood/recall. Population-level JSD alone is not enough.
3. Report uncertainty estimates across random seeds, cities, or bootstrap resamples for Table 1-4.
4. Include an ablation on the teacher-generated SFT data and LLM-as-judge filtering. For example, how much performance drops if no teacher reasoning data are used, or if hard cases are retained.

5. Provide exact prompts, decoding parameters, model checkpoints, API/model versions, and tool settings for all LLM baselines.
6. Expand the ethical discussion to include re-identification risks, possible misuse of synthetic mobility traces, and concrete safeguards for data/software release.
7. Clarify whether the synthesized user profiles are provided to all LLM baselines in the same format as MobiWeaver, and whether deep learning baselines have access to equivalent user information.
8. The paper should avoid overclaiming cross-city generalization unless no target-city training information is used beyond the allowed retrieval database and prior user histories. The current setting still uses held-out-city training splits to instantiate retrieval tools (Sec. 3.1).

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 2.5
Excitement: 2.5
Overall Assessment: 2.5 = Borderline Findings
Ethical Concerns:
There are no concerns with this submission.

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.
Datasets: 3 = Potentially useful: Someone might find the new datasets useful for their work.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
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Add: **Official Comment**

Response to Reviewer uuCd (1/4)

Official Comment by Authors 13 Jul 2026, 21:47

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uuCd

Comment:
Thank you for your detailed and insightful review. We appreciate your thorough engagement with our method. If you are not satisfied with our answers or have more questions, please let us know as soon as possible, so that we can try our best to answer any further questions before the deadline.

> Weakness 1: The Main Empirical Comparison Is Not Fully Convincing

Thank you for raising this important concern. We respond from two perspectives: the baseline setup and the ablation studies.

- **The mobility generation baselines themselves only provide prompting strategies and workflow harnesses, without any corresponding training procedure.** To faithfully reproduce these methods, we evaluate them under their official settings rather than introducing additional optimization. At inference time, all LLM-based baselines are given access to the same external information as MobiWeaver (Lines 343–351). **Artificially adding new optimization strategies would produce newly adapted methods rather than the original baselines.**
- **Tables 2 and 4 provide ablations that separately verify the effectiveness of the hierarchical architecture and TACPO.** In Table 2, the **w/o Hierarchy** variant is trained with the same training data and strategy, but replaces the planner–searcher architecture with a single 8B model. It

achieves clearly worse performance than MobiWeaver. **This shows that the gain does not merely come from teacher supervision or RL training, but also from the planner–searcher decomposition.** Similarly, in Table 4, adding either Effectiveness-Aware Rescaling or Position-Aware Rescaling further improves performance. The complete TACPO achieves the best results, supporting the contribution of both components.

Weakness 2 & Suggestion 2: Insufficient Individual-Level Evaluation

Thank you for this valuable suggestion. We agree that adding individual-level metrics provides a more comprehensive evaluation of MobiWeaver. **Following your suggestion, we additionally evaluate individual-level trajectory fidelity from both temporal and spatial perspectives.**

- **Temporal fidelity.** We calculate the **Start–End Time Error (SETE)** to measure the deviation between the predicted and real daily activity windows. For a ground-truth trajectory with activity window $[t^\star_{\text{start}}, t^\star_{\text{end}}]$ and a generated trajectory with activity window $[t_{\text{start}}, t_{\text{end}}]$, the trajectory-level SETE is defined as: $\text{SETE} = \frac{|t_{\text{start}} - t^\star_{\text{start}}| + |t_{\text{end}} - t^\star_{\text{end}}|}{2}$. All timestamps are converted into minutes.

Model	Melbourne	Sydney	Tokyo
DeepSeek-V3.2	327.80	331.94	263.64
Qwen3.5-397B-A17B	190.63	200.99	188.32
GPT-OSS-120B	278.14	294.53	216.28
Qwen3.5-9B	224.12	313.21	238.34
MobiWeaver	**104.39**	**115.94**	**116.60**

- **Spatial fidelity.** We calculate the **exact POI hit rate** to measure whether the model recovers the concrete POIs in each user's trajectory.

Model	Melbourne	Sydney	Tokyo
DeepSeek-V3.2	11.77	12.80	11.86
Qwen3.5-397B-A17B	15.42	17.35	18.56
GPT-OSS-120B	13.39	15.91	16.45
Qwen3.5-9B	8.01	10.42	10.53
MobiWeaver	**26.92**	**29.56**	**32.84**

- **We primarily report population-level JSD metrics to follow established protocols [1,2] and maintain consistency with prior work**. We will add a discussion of individual-level evaluation in the revised manuscript.

Add: **Official Comment**

Response to Reviewer uuCd (2/4)

Official Comment by Authors 13 Jul 2026, 21:48

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uuCd

Comment:

Weakness 3 & Suggestion 4: Potential Bias from Teacher-Based Data Filtering

Thank you for raising this concern. We respond from two perspectives: the optimization noise introduced by overly difficult samples and an additional ablation experiment.

- **The main purpose of filtering is to prevent samples that cannot provide effective learning signals during RL training.** The teacher model has substantially stronger reasoning capability than the base policy. During RL training, the smaller base policy is more likely to produce rollout groups with all-zero or nearly identical rewards on samples that even the teacher model struggles to solve. Such groups provide little useful advantage signal and may destabilize optimization. **Filtering samples that are beyond the current policy's learning capability is also a common practice in recent RLVR training [3,4].**

- **Following your suggestion, we train MobiWeaver without teacher-based filtering.** Specifically, we retain the samples that are rejected during teacher synthesis, while keeping the remaining training settings unchanged. **Removing teacher filtering consistently degrades performance across all five metrics.**

Variant	SD	SI	DARD	STVD	ActP
w/o Teacher Filtering	0.0185	0.0396	0.3209	0.5287	0.0495
MobiWeaver	0.0173	0.0372	0.3140	0.5248	0.0476

Weakness 4: Limited Reproducibility

Thank you for pointing this out. We agree that the data synthesis pipeline and the resulting training resources should be fully reproducible. **To preserve anonymity during the review process, we have not yet publicly released these resources.** Upon completion of the review process, we will release the complete synthesis pipeline, the generated training datasets, and the constructed city databases.

Weakness 5, Suggestion 5 and Suggestion 7: Underspecified Baseline Configuration

Thank you for pointing out this reproducibility issue. We respond from the following three aspects.

- **For general LLMs, we use the model versions cited in Table 1. All models are evaluated under the same inference configuration.** During evaluation, we set the temperature to 0.6, top-p to 0.95, top-k to -1, and max_tokens to 32K. **As described in Lines 339–343 and 1039–1041**, all general LLMs use the MobiWeaver tool set except for the planner-specific `get_plan` tool. Their prompts are also aligned as closely as possible with the MobiWeaver-Searcher prompt, with only the instructions for invoking `get_plan` removed (Line 1035-1039).
- **Mobility LLMs are evaluated using their official prompts and tool sets, with GPT-OSS-120B as the common backbone.** As described in Lines 343–347 and 1041–1043, we follow the prompt templates, workflows, and tool definitions provided by their official implementations. Their decoding settings are also unified, with a temperature of 0.6, top-p of 0.95, top-k of -1, and max_tokens of 32K.
- **To ensure a fair comparison, all LLM baselines and MobiWeaver have access to the same user profiles, historical trajectories, and POI information, as described in Lines 348–351.** Due to the input limitations of the deep learning baselines, they use only historical trajectories during inference.

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Replying to Response to Reviewer uuCd (2/4)

Response to Reviewer uuCd (3/4)

Official Comment by Authors 13 Jul 2026, 21:49

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uuCd

Comment:

Weakness 6 & Suggestion 3: Lack of Uncertainty Estimates

Thank you for pointing out this issue. We agree that uncertainty estimates make the ablation results more statistically reliable and easier to interpret. **Following your suggestion, we conduct trajectory-level bootstrap resampling over the evaluation set.**

Specifically, in each bootstrap iteration, we independently sample trajectories with replacement within the test sets of Melbourne, Sydney, and Tokyo until reaching the original number of test trajectories in each city. We then reconstruct the corresponding trajectory distributions and recompute the five metrics for each city. We repeat this procedure 1000 times and construct 95% confidence intervals using the 2.5th and 97.5th percentiles of the bootstrap distribution. The revised Tables 2–4 are shown below.

Variant	SD	SI	DARD	STVD	ActP
r/ planner	0.0408 [0.0393, 0.0424]	0.0871 [0.0845, 0.0900]	0.5111 [0.4996, 0.5238]	0.6789 [0.6688, 0.6900]	0.1427 [0.1379, 0.1480]
r/ searcher	0.0311 [0.0298, 0.0325]	0.0414 [0.0396, 0.0434]	0.4365 [0.4258, 0.4483]	0.6013 [0.5919, 0.6117]	0.0722 [0.0691, 0.0757]
w/o Hierarchy	0.0214 [0.0203, 0.0226]	0.0460 [0.0441, 0.0481]	0.3782 [0.3682, 0.3893]	0.5712 [0.5620, 0.5813]	0.0644 [0.0614, 0.0677]
MobiWeaver	**0.0173 [0.0163, 0.0185]**	**0.0372 [0.0354, 0.0392]**	**0.3140 [0.3047, 0.3243]**	**0.5248 [0.5160, 0.5345]**	**0.0476 [0.0450, 0.0504]**

Variant	SD	SI	DARD	STVD	ActP
w/o r_{tp}^{P}	0.0196 [0.0185, 0.0208]	0.0407 [0.0389, 0.0427]	0.3314 [0.3219, 0.3419]	0.5392 [0.5303, 0.5490]	0.0508 [0.0482, 0.0537]
w/o r_{cat}^{P}	0.0231 [0.0219, 0.0244]	0.0382 [0.0364, 0.0402]	0.3666 [0.3567, 0.3775]	0.5520 [0.5430, 0.5619]	0.0571 [0.0543, 0.0602]
w/o r_{st}^{P}	0.0283 [0.0270, 0.0297]	0.0396 [0.0378, 0.0416]	0.4092 [0.3988, 0.4207]	0.6155 [0.6060, 0.6260]	0.0985 [0.0947, 0.1027]
MobiWeaver	**0.0173 [0.0163, 0.0185]**	**0.0372 [0.0354, 0.0392]**	**0.3140 [0.3047, 0.3243]**	**0.5248 [0.5160, 0.5345]**	**0.0476 [0.0450, 0.0504]**

Variant	SD	SI	DARD	STVD	ActP
Qwen3-4B-Instruct	0.0311 [0.0298, 0.0325]	0.0414 [0.0396, 0.0434]	0.4365 [0.4258, 0.4483]	0.6013 [0.5919, 0.6117]	0.0722 [0.0691, 0.0757]
+GRPO	0.0269 [0.0257, 0.0283]	0.0392 [0.0374, 0.0412]	0.3743 [0.3643, 0.3853]	0.5707 [0.5615, 0.5808]	0.0582 [0.0554, 0.0613]
+TACPO w/o PA	0.0189 [0.0178, 0.0201]	0.0376 [0.0358, 0.0396]	0.3361 [0.3266, 0.3466]	0.5396 [0.5307, 0.5494]	0.0502 [0.0476, 0.0531]
+TACPO w/o EA	0.0196 [0.0185, 0.0208]	0.0380 [0.0362, 0.0400]	0.3282 [0.3188, 0.3386]	0.5552 [0.5462, 0.5652]	0.0514 [0.0488, 0.0543]

Variant	SD	SI	DARD	STVD	ActP
MobiWeaver	**0.0173 [0.0163, 0.0185]**	**0.0372 [0.0354, 0.0392]**	**0.3140 [0.3047, 0.3243]**	**0.5248 [0.5160, 0.5345]**	**0.0476 [0.0450, 0.0504]**

The results preserve the original overall rankings and confirm the contribution of each component.

> Weakness 7 & Suggestion 6: Insufficient Ethical Discussion

Thank you for pointing this out. We will expand the Ethical Considerations section from the following three aspects to more clearly discuss the privacy risks.

- **MobiWeaver does not access any direct identity information, and all trajectories are obtained from a de-identified public benchmark.** The Massive-STEPS providers removed private information. MobiWeaver only uses the anonymized user IDs and trajectory records and has no access to any external mapping that could recover the users' real identities. This substantially reduces direct re-identification risk.
- **The synthesized user profiles mainly summarize users' mobility regularities.** As shown in Table 8, the profile fields mainly describe mobility patterns. They are automatically inferred from anonymized trajectories and should not be interpreted as verified personal information about real individuals. **Even so, we will restrict the release of user-level profiles and avoid publishing synthesized profiles and generated trajectories.**
- **We will introduce explicit safeguards for the release and use of the code and data.** The data and software licenses, repository documentation, and usage statement will explicitly prohibit using MobiWeaver or its generated traces for personal tracking, identity inference, surveillance, or other applications targeting real individuals.

Add: **Official Comment**

Replying to Response to Reviewer uuCd (3/4)

Response to Reviewer uuCd (4/4)

Official Comment by Authors 13 Jul 2026, 21:50

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uuCd

Comment:

> Weakness 8: Indirect NLP Contribution

Thank you for pointing this out. We respond from the following two perspectives.

- **TACPO addresses a general credit-assignment problem in LLM agentic RL.** In long-horizon trajectories, effective and ineffective actions may coexist within the same rollout, while vanilla GRPO assigns them the same rollout-level advantage. TACPO introduces Effectiveness-Aware Rescaling to adjust the advantages of specific action-token spans according to their local effectiveness, enabling more fine-grained credit assignment. This idea is applicable to other multi-step agents.
- **Urban computing is an important application domain for NLP.** MobiWeaver demonstrates how natural-language user profiles and historical trajectories can be used to synthesize mobility trajectories, providing a approach and a concrete example for applying NLP to urban computing.

> Suggestion 1: Add a Fairness Table

Thank you for raising this point, which is closely related to Weaknesses 1 and 5.

- **All LLM baselines and MobiWeaver have access to the same tools and external information,as described in Lines 348–351.** To faithfully reproduce the baselines, we follow their official prompts, workflows, and inference harnesses without introducing additional training data and adaptation.
- **All LLM baselines use the same decoding configuration.** We set the temperature to 0.6, top-p to 0.95, top-k to -1, and max_tokens to 32K. We will add this setting to the revised paper.

> Suggestion 8: Clarify the Scope of Cross-City Generalization

Thank you for pointing this out.

- **We clarify that MobiWeaver does not use any target-city training information beyond the permitted retrieval database and temporally prior user histories.** The held-out training split is mainly used to extract static POI metadata and construct the POI database. This setting directly corresponds to our core transfer objective in Lines 111–112: Transfer to a new city using its database, without updating parameters.
- **We further conduct a history-free zero-shot transfer experiment.** Specifically, we force all interfaces that retrieve user historical trajectories to return empty results and reevaluate all models without access to any target-user trajectory records. Therefore, **the official target-city training split is used only to construct the POI database and provides NO auxiliary trajectories**. In real-world cold-start deployment, the same static POI information can be obtained from map APIs.

Model	SD	SI	DARD	STVD	ActP
DeepSeek-V3.2	0.0502	0.1218	0.6516	0.7724	0.1789
Qwen3.5-397B-A17B	0.0331	0.0616	0.4827	0.6629	0.1195
GPT-OSS-120B	0.0507	0.1013	0.6638	0.7569	0.2476
Qwen3.5-9B	0.0864	0.0967	0.8462	0.7928	0.3921
MobiWeaver	**0.0281**	**0.0511**	**0.4018**	**0.5977**	**0.0784**

MobiWeaver still achieves the best performance across all five metrics without access to user historical trajectories. We will add this experiment and clarification to the revised manuscript.

Reference

[1] Large language models as urban residents: An llm agent framework for personal mobility generation. NIPS, 2024

[2] Chain-of-planned-behaviour workflow elicits few-shot mobility generation in llms. 2024.

[3] MARS2: Scaling Multi-Agent Tree Search via Reinforcement Learning for Code Generation. ACL, 2026

[4] UR^2: Unify RAG and Reasoning through Reinforcement Learning. ACL, 2026

Add: Official Comment

Official Review of Submission8297 by Reviewer xd7Z

Official Review by Reviewer xd7Z 29 Jun 2026, 18:24 (modified: 12 Jul 2026, 23:34)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer xd7Z

Revisions (/revisions?id=3pOLP0qk4s)

Paper Summary:

This paper proposes a hierarchical multi-agent framework that separates cross-city mobility generation task into a planner, which generates coarse time and activity itineraries from user profiles and historical behavior, and a tool-augmented searcher, which retrieves local POIs and grounds the itinerary into timestamped trajectories using an external city database. To improve reinforcement-learning training for the searcher, the authors introduce Tool-Anchored Credit Policy Optimization (TACPO), which assigns credit specifically to POI-related action tokens, downweights ineffective tool-use decisions, and gives greater importance to early POI selections that influence subsequent steps. Experiments on three cities show that MobiWeaver achieves the lowest JSD metrics.

Summary Of Strengths:
S1. The paper addresses cross-city mobility generation, which is challenging and meaningful. The proposed method enables adaptation to a new city by replacing the external mobility database rather than retraining city-specific parameters, which is a useful direction for application.

S2. The hierarchical decomposition is well motivated and looks reasonable. Separating the task into a city-agnostic planner for coarse time–category itineraries and a tool-augmented searcher for concrete POI grounding matches the two distinct requirements of mobility generation: high-level behavioral reasoning and local spatial retrieval. This design also provides a mechanism for reducing interference between intent modeling and POI-level grounding.

S3. The reported results are consistently strong. MobiWeaver achieves the best results across all five metrics in Melbourne, Sydney, and Tokyo, against compared baseline models.

Summary Of Weaknesses:
W1. Although MobiWeaver does not fine-tune its model weights in a new city, it still uses that official training split of the target city to build the POI database and provide user histories. Therefore, the transfer setting assumes the data availability of a target city and may not reflect truly cold-start deployment in a new city with sparse, incomplete, or unavailable mobility records.

W2. The presentation and motivation could be improved. In particular, the connection between the policy-level and action-level optimization challenges and the specific mobility-generation task is not sufficiently intuitive in the introduction. The discussion introduces these two optimization issues, but I can hardly align these issues with the mobility generation task. Moreover, Figure 1 is not very informative and does not effectively support this motivation.

W3. Reproducibility is a marjor concern as the source code and relevant data resources are not available. In addition, the paper acknowledges that its tool interface relies on a self-constructed city database rather than publicly available map-provider APIs. Since POI retrieval is central to the proposed searcher, the results may depend substantially on the coverage, quality, and freshness of this database.

Comments Suggestions And Typos:
The motivation provided in intro section can be improved further.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 3.5
Excitement: 3.5
Overall Assessment: 3.5 = Borderline Conference
Limitations And Societal Impact:
Yes.

Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

Add: **Official Comment**

Response to Reviewer xd7Z

Official Comment by Authors 13 Jul 2026, 19:56

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer xd7Z

Comment:

Thank you for your great efforts on the review and constructive comments. We will try our best to answer all your questions. If you are not satisfied with our answers or have more questions, please let us know as soon as possible, so that we can try our best to answer any further questions before the deadline.

> Weakness 1: Dependence on Target-City Data and User Histories

Thank you for this valuable comment. **To more rigorously validate MobiWeaver's cross-city transfer capability, we conduct a history-free zero-shot transfer experiment.**

Specifically, we force all interfaces that retrieve user historical trajectories to return empty results and reevaluate all models without access to any target-user trajectory records. Therefore, **the official target-city training split is used only to construct the POI database and provides NO auxiliary trajectories**. In real-world cold-start deployment, the same static POI information can be obtained from map APIs.

Model	SD	SI	DARD	STVD	ActP
DeepSeek-V3.2	0.0502	0.1218	0.6516	0.7724	0.1789
Qwen3.5-397B-A17B	0.0331	0.0616	0.4827	0.6629	0.1195
GPT-OSS-120B	0.0507	0.1013	0.6638	0.7569	0.2476
Qwen3.5-9B	0.0864	0.0967	0.8462	0.7928	0.3921
MobiWeaver	**0.0281**	**0.0511**	**0.4018**	**0.5977**	**0.0784**

MobiWeaver achieves the best performance across all five metrics without access to user historical trajectories. This demonstrates that its cross-city transferability does not rely solely on retrieving mobility records. We will add this experiment and clarification to the revised manuscript.

> Weakness 2: Insufficiently Intuitive Motivation and Figure 1

Thank you for this valuable comment. We agree that the current introduction could be made more intuitive. **We will add a concrete trajectory case and revise Figure 1 to make the motivation clearer.**

Suppose the ground-truth trajectory is **Office Building O1 → Restaurant R1 → Shopping Mall S1**.

- **Positive-advantage rollouts may contain ineffective searches, while negative-advantage rollouts may contain useful ones (Lines 80–82).** For example, the model correctly select O1 and R1 but incorrectly select S2. The rollout may still receive a positive advantage, and vanilla GRPO assigns this positive signal to all intermediate actions, **including the incorrect selection of S2**. Conversely, a low-scoring rollout may still contain a correct selection such as O1, **but this correct action is also penalized**. These cases correspond to the positive- and negative-rollout curves in the top panel of Figure 1.

- **Early POI anchors have a larger impact on later trajectory quality (Lines 82–83).** Mobility trajectories are generated autoregressively, and each selected POI provides a spatial anchor for subsequent retrieval. If the model initially selects an incorrect office building O2 that is **far from O1**, later searches are performed around the wrong region, **making it difficult to retrieve the correct restaurant R1 and shopping mall S1**. In contrast, selecting O1 correctly provides an appropriate spatial constraint for subsequent generation. This is supported by the bottom panel of Figure 1, where inserting the correct initial POIs improves the quality of the remaining trajectory.

> Weakness 3: Reproducibility of the Code and Data

Thank you for pointing this out. We agree that the data synthesis code and the resulting training resources should be fully reproducible. To preserve anonymity during the review process, we have not yet publicly released these resources. **After the review process, we will release the complete synthesis pipeline, the generated training dataset and databases.**

Add: **Official Comment**

Reviewer Checklist of Submission8297 by Reviewer xd7Z

Reviewer Checklist by Reviewer xd7Z 29 Jun 2026, 18:22

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers

Appropriateness: Yes
Appropriateness Justification: N/A - this paper makes a contribution to computational processing of natural language.
Formatting: Yes
Formatting Justification: N/A - this paper is properly formatted.
Length: Yes
Length Justification: N/A - this paper is within the page limits.
Anonymity: Yes
Anonymity Justification: N/A - this paper is properly anonymized.
Limitations: Yes
Limitations Justification: N/A - this paper has the 'Limitations' section.
Overall Level: Yes
Overall Level Justification: N/A - this seems like a good-faith submission worthy of full review.
Responsible Checklist: Yes
Need Ethics Review: No
Ethics Review Justification: N/A - this paper does not need an ethics review.

Add: **Official Comment**

Official Review of Submission8297 by Reviewer uX97

Official Review by Reviewer uX97 26 Jun 2026, 14:04 (modified: 12 Jul 2026, 23:34)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uX97

Revisions (/revisions?id=kugTeKzHXV)

Paper Summary:
Authors developed "MobiWeaver," a hierarchical multi-agent framework consisting of a planner and a searcher, to realistically generate human mobility trajectories (POI trajectories) in cross-city scenarios.

Summary Of Strengths:
S1: The hierarchical design offloads city-specific spatial knowledge to an external database, enabling cross-city transfer without the need for parameter updates.

S2: It is highly impressive that a relatively small model with 8B parameters achieves performance surpassing that of LLMs with hundreds of billions of parameters.

Summary Of Weaknesses:
W1: From a practical perspective, because the process involves multiple tool calls between agents, the inference cost (in terms of token consumption and time) required to generate a single day's trajectory for one individual is extremely heavy.

W2: Human mobility behavior is strongly influenced not only by individual demographic attributes and past tendencies (daily routines) but also by "dynamic external contexts," such as the day's weather (e.g., travel restrictions due to rain), special social events in the city (e.g., festivals, sports matches), or specific holiday occurrences. However, the MobiWeaver planner bases its predictions solely on static user profiles, past temporal history, and basic time metadata like the day of the week. As a result, the model is excessively biased toward mimicking "average, daily routines" and fails to reproduce the flexible changes in human behavior (anomalies) that occur in response to sudden events or environmental shifts.

W3: There seems to be a significant risk of introducing social stereotypes during the artificial synthesis of user profiles. Consequently, the trained model may inherit these issues, making it necessary to consider the critical risk of a feedback loop where social biases and flawed causal relationships become "reinforced" (hardcoded) into the model itself.

W4: Because the hyperparameter settings for TACPO (such as rescaling weights) are delicate, they may be overly tuned (overfitted) to the source city. This raises the concern that, ultimately, hyperparameter readjustment and retraining might still be required for each new target city.

Comments Suggestions And Typos:
N/A

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.
Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.
Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.
Best Paper Justification:
N/A

Limitations And Societal Impact:
N/A

Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
Datasets: 3 = Potentially useful: Someone might find the new datasets useful for their work.
Software: 3 = Potentially useful: Someone might find the new software useful for their work.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Add: **Official Comment**

Response to Reviewer uX97 (1/2)

Official Comment by Authors 13 Jul 2026, 20:01

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uX97

Comment:

Thank you for your careful review and comments. We address each concern below. If any point remains unclear or you have further questions, please let us know before the deadline so that we can respond promptly.

Weakness 1: Inference Cost

Thank you for pointing this out. We agree that inference-cost analysis is necessary for a more comprehensive evaluation of MobiWeaver. **Following your suggestion, we evaluate both token consumption and inference time**. We will add this analysis to the revised paper.

- **MobiWeaver consumes fewer tokens than the baselines.** We report rollout length in the format `mean [P50, P90]`. For MobiWeaver, the token count includes the combined outputs of both the Planner and Searcher. Across the three cities, MobiWeaver averages **7468 tokens per trajectory**, compared with **8655 tokens** for the second-lowest model.

Model	Melbourne	Sydney	Tokyo
DeepSeek-V3.2	9971 [9846, 13304]	9939 [9889, 13168]	8772 [8588, 12169]
Qwen3.5-397B-A17B	13284 [11881, 25065]	13339 [11898, 23192]	13185 [11960, 24593]
GPT-OSS-120B	8864 [8729, 13556]	9679 [9522, 13230]	7423 [7425, 11136]
Gemma4-26B-A4B	11693 [8201, 24260]	12722 [8999, 24617]	18249 [14766, 35128]
Qwen3.5-9B	10983 [10833, 15943]	10957 [10845, 15691]	9909 [9673, 14670]
LLMob	11150 [10593, 16035]	11919 [11737, 16532]	9722 [9645, 13478]
CoPB	10248 [10104, 14981]	11131 [10978, 15306]	8807 [8759, 12554]
MobiWeaver	**7798 [7651, 10312]**	**8115 [8002, 10667]**	**6491 [6324, 8762]**

- **MobiWeaver also requires less inference time than similarly sized baselines.** We deploy all locally evaluated models on the same two H200 GPUs and report the total end-to-end inference time in minutes. Larger baselines are accessed through external APIs and therefore cannot be compared fairly under identical hardware. **Nevertheless, their longer rollouts and larger parameters suggest that their latency would generally be higher.**

Model	Melbourne	Sydney	Tokyo
Gemma4-26B-A4B	86.27	118.92	63.88
Qwen3.5-9B	124.95	153.14	52.36
MobiWeaver	**48.13**	**69.39**	**17.25**

Weakness 2: Missing Dynamic Context

Thank you for raising this concern. We respond from three aspects.

- **We directly evaluate whether MobiWeaver merely reproduces "average daily routines" on non-routine trajectories.**
  1. We serialize each evaluation trajectory, including its timestamps, POIs, and coordinates, and encode it with Qwen3-Embedding-8B to obtain a trajectory representation $\mathbf{z_{u,i}}$.
  2. For each user u, we compute the cosine distance between every pair of trajectories: $d_{ij} = 1 - \frac{\mathbf{z_{u,i}}^\top \mathbf{z_{u,j}}}{|\mathbf{z_{u,i}}||\mathbf{z_{u,j}}|}$.
  3. For each trajectory, we calculate its average distance from the user's other trajectories: $A_{u,i} = \frac{1}{n_u - 1} \sum_{j \neq i} d_{ij}$.
  4. We select the top 20% trajectories with the largest $A_{u,i}$ for each user as the **non-routine subset** and reevaluate models.

Model	SD	SI	DARD	STVD	ActP
DeepSeek-V3.2	0.0431	0.1162	0.6148	0.7024	0.1412
Qwen3.5-397B-A17B	0.0312	0.0614	0.4821	0.6733	0.1022
Qwen3.5-9B	0.0622	0.0719	0.7194	0.7118	0.2652
MobiWeaver	0.0233	0.0508	0.3978	0.6052	0.0614

MobiWeaver remains best on all five metrics. **This shows that MobiWeaver does not merely imitate average daily routines.**

- **Modeling non-routine behavior caused by sudden events is a distinct and challenging problem in spatiotemporal modeling.** Several studies focus on this issue in time-series modeling [1,2] and traffic forecasting [3,4]. In human mobility generation, most prior studies focus on routine mobility patterns [5]. ELLMob [6] recently formulates event-driven mobility generation as a separate task and constructs a dedicated event-centric dataset. **In contrast, MobiWeaver focuses on combining transferable user mobility patterns with target-city POI knowledge for cross-city generation**. We will clarify this task boundary in the revised paper.
- **Missing event information.** As stated in Lines 597–602, Massive-STEPS does not provide event annotations. To our knowledge, other common datasets, such as Tencent and ChinaMobile [7], likewise do not provide external events. ELLMob treats the construction of such an event-centric dataset as one of its main contributions, while its resources have not yet been fully released.

Add: **Official Comment**

Response to Reviewer uX97 (2/2)

Official Comment by Authors 13 Jul 2026, 20:02

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uX97

Comment:

> **Weakness 3: Potential Social Stereotypes and Feedback Loops in Synthetic User Profiles**

Thank you for raising this concern. We respond from the profile-construction method, training process, and prior works.

- **The synthesized user profiles mainly summarize users' mobility regularities rather than characterizing users' real private identities, which limits the risk of introducing social stereotypes.** As shown in Table 8, the profile mainly describes long-term mobility characteristics. These fields are synthesized from historical trajectories and do not contain real-world private information, as clarified in Ethical Considerations (Lines 603–615).
- **The MobiWeaver training pipeline does not create the feedback loop.** User profiles are generated once during data preprocessing and remain fixed throughout training. Generated trajectories are never used to regenerate profiles. Thus, there is no recursive process in which biased generations are repeatedly fed back and reinforced.
- **Our construction procedure follows prior works to ensure fair evaluation.** As described in Lines 882–885, because public mobility datasets do not provide user profiles, we follow prior work and summarize users' mobility patterns from their historical trajectories.

> **Weakness 4: Potential Target-City Overfitting of TACPO Hyperparameters**

Thank you for raising this concern. We clarify that **avoiding target-city retraining is precisely a key objective and advantage of MobiWeaver**. We respond from the perspectives of model training and experimental evaluation.

- **MobiWeaver is not trained on a single source city, but learns transferable mobility intentions from diverse cities.** As shown in Figure 6 and described in Lines 299–315, MobiWeaver is trained on 58K trajectories from 10 cities. This multi-city training enables the model to learn general mobility intentions. When deployed in a new city, MobiWeaver only replaces the external mobility database and requires no parameter update（Lines 152–157）.
- **None of our experiments involves city-specific fine-tuning.（Lines 317–323）** Under this setting, MobiWeaver achieves state-of-the-art performance across all three held-out cities without retraining.

Reference

[1] APT: Affine prototype-timestamp for time series forecasting under distribution shift. AAAI, 2026.

[2] Adaptive conformal anomaly detection with time series foundation models for signal monitoring. ICLR, 2026.

[3] Hyperd: Hybrid periodicity decoupling framework for traffic forecasting. AAAI 2026.

[4] A Two-Stage Anomaly-Aware Framework for Robust Traffic Forecasting with Memory-Guided GNNs. WSDM, 2026

[5] Large language models as urban residents: An llm agent framework for personal mobility generation. NIPS, 2024

[6] ELLMob: Event-Driven Human Mobility Generation with Self-Aligned LLM Framework. ICLR 2026

[7] Chain-of-planned-behaviour workflow elicits few-shot mobility generation in llms. 2024.

Add: **Official Comment**

Action Editor Checklist by Area Chair HxEa

Action Editor Checklist by Area Chair HxEa (Yuxuan Liang (/profile?id=~Yuxuan_Liang1))

22 Jun 2026, 11:36 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers

Appropriateness: Yes
Appropriateness Justification: N/A - this paper makes a contribution to computational processing of natural language.
Formatting: Yes
Formatting Justification: N/A - this paper is properly formatted.
Length: Yes
Length Justification: N/A - this paper is within the page limits.
Anonymity: Yes
Anonymity Justification: N/A - this paper is properly anonymized.
Limitations: Yes
Limitations Justification: N/A - this paper has the 'Limitations' section.
Overall Level: Yes
Overall Level Justification: N/A - this seems like a good-faith submission worthy of full review.
Responsible Checklist: Yes
Need Ethics Review: No
Ethics Review Justification: N/A - this paper does not need an ethics review.
Number Of Assignments: Yes
Diversity: Yes
Resubmission: This paper is a not a resubmission
Resubmission Reassignments: This paper is a not a resubmission
Resubmission Notes: This paper is a not a resubmission

Add: **Official Comment**